

Marcus Chen I WADA

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EDUCATION

University of Michigan–Dearborn, Dearborn, MI

Expected May 2027

B.S. in Computer Engineering

GPA: 3.94/4.00

Relevant Coursework: Applied Linear Algebra (MATH 423); Advanced Topics in Power System (ECE 616, audited)

RESEARCH INTERESTS

Statistical learning and predictive modeling for reliable, resilient and equitable energy infrastructure planning; optimization and control of energy systems; scalable algorithm design using high-performance computing for large-scale power system applications; and real time simulation and hardware-in-the-loop validation

PUBLICATIONS

Journal Articles

- [1] S. Mohammadi, W. Wang, **M. C. I. Wada**, R. Haghighi, A. Hassan, H. Liu, A. Bhatnagar, A. Chen, and W. Su, “Grid integration of AI data centers: A critical review of energy storage solutions,” *Advances in Applied Energy*, vol. 23, Art. no. 100291, 2026. [doi:10.1016/j.adapen.2026.100291](https://doi.org/10.1016/j.adapen.2026.100291)

Conference Papers

- [1] R. Haghighi, A. Hassan, S. Mohammadi, **M. C. I. Wada**, and W. Su, “A Stackelberg–Bayesian capacity-market game of carbon regulation and second-life battery investment under AI data-center load growth,” accepted for presentation at the *2026 North American Power Symposium (NAPS)*, 2026

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Aug. 2023–Present

University of Michigan–Dearborn

Dearborn, Michigan

Advisor: Prof. Wencong Su

- Developed data collection and processing pipelines for utility outage records; trained LSTM-based forecasting models and integrated predictive and analytical capabilities into an LLM assisted dashboard
- Designed an optimization based converter selection tool for microgrid applications for a Toyota sponsored project
- Evaluated hierarchical energy-storage architectures for AI data centers and proposed strategies for load smoothing, renewable integration, and improved grid compatibility
- Contributed to the formulation of a Bayesian Stackelberg–Nash model for evaluating carbon regulation, capacity-market decisions, and second-life battery investment under AI data-center load growth

INDUSTRY EXPERIENCE

Controls and Energy Systems Engineer

Jun. 2026–Present

GC and Mobility

Remote

- Develop QP models for energy management and coordinated charging of neighborhood electric-vehicle fleets, with evaluation through microgrid simulation and hardware-in-the-loop testing

SELECTED ENGINEERING PROJECTS

Closed Loop Airbrake Controller for Sounding Rockets

Aug. 2024–Present

Michigan Aeronautical Science Association at Dearborn

- Led the development of the team's **first successfully deployed** embedded airbrake system for the International Rocket Engineering Competition
- Implemented and tuned Kalman and Madgwick filter based state estimation, real-time apogee prediction, and closed-loop guidance algorithms

PRESENTATIONS AND POSTERS

- **M. C. I. Wada**, W. Su, and L. Andres, "A Comprehensive Python Package for Forecasting Power Outages," poster presented at the Summer Undergraduate Research Experience (**SURE**) Program, University of Michigan–Dearborn, Dearborn, Michigan, Sept. 17, 2024. [doi:10.7302/24853](https://doi.org/10.7302/24853)

TEACHING EXPERIENCE

MATH 116: Calculus II

University of Michigan–Dearborn

Winter 2024, Fall 2025

Supplemental Instruction Leader

- Led collaborative problem-solving and examination-review sessions, developed instructional materials, and provided tutoring through the Mathematics Learning Center

CHEM 134: General Chemistry

University of Michigan–Dearborn

Fall 2024

Supplemental Instruction Leader

- Led weekly peer-learning sessions, developed practice problems and review materials, and supported students during in-class assignments

HONORS AND AWARDS

Undergraduate

- **James B. Angell Scholar**, University of Michigan, 2024–2026
- **Fourth Place**, Lower Michigan Mathematics Competition, 2025
- **Undergraduate International Scholarship**, University of Michigan–Dearborn, 2023–2027

High School

- **Four-Time Gold Medalist, National Finalist**, Brazilian Astronomy Olympiad, 2019–2022
- **Two-Time Gold Medalist**, Brazilian National Science Olympiad, 2021–2022
- **Honorable Mention**, Brazilian Mathematics Olympiad, 2019
- **Bronze Medal**, Brazilian Economics Olympiad, 2022

SKILLS

Computing: Python, MATLAB/Simulink, embedded C/C++, and HPC (SLURM, MPI)

Languages: Portuguese (native); English (fluent); Mandarin Chinese (conversational)

LEADERSHIP AND SERVICE

Treasurer

Mathematics Club, University of Michigan–Dearborn

Aug. 2023–Present

Dearborn, Michigan

- Secured more than **\$2,000** in university funding and supported competitions, colloquia, workshops, and Putnam preparation activities

Corresponding Secretary

IEEE–Eta Kappa Nu, Theta Tau Chapter

2025–2026

Dearborn, Michigan

- Organized chapter activities and represented the chapter at the IEEE–HKN Student Leadership Conference